21 $p^3 \times p^x = p^9$

(a) Find the value of x.

Multiplying powers with the same base: add exponents.

$3 + x = 9$, so $x = 9 - 3 = 6$.

$x =$ 6

1 mark
Correct exponent

(1)

$(7^2)^y = 7^{10}$

(b) Find the value of y.

Raising a power to a power: multiply exponents.

$2y = 10$, so $y = 10 \div 2 = 5$.

$y =$ 5

1 mark
Correct exponent

(1)

$100^a \times 1000^b$ can be written in the form 10^w

(c) Show that $w = 2a + 3b$

$100 = 10^2$ and $1000 = 10^3$

1 mark
Write each base as a power of 10

$100^a \times 1000^b = (10^2)^a \times (10^3)^b$

$= 10^{2a} \times 10^{3b} = 10^{2a + 3b}$ so $w = 2a + 3b$.

1 mark
Multiply exponents, then add them to complete the proof

(2)

(Total for Question 21 is 4 marks)